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A Comprehensive Review on Electrical Discharge Machining and Wire Electrical Discharge Machining of Titanium Alloys

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ABSTRACT

Titanium alloys are extensively employed in aerospace, automotive, biomedical, and defense industries due to their superior strength-to-weight ratio, corrosion resistance, and high-temperature performance. However, their poor machinability using conventional machining methods poses significant challenges such as excessive tool wear, poor surface integrity, and low productivity. Electrical Discharge Machining (EDM) and Wire Electrical Discharge Machining (WEDM) have emerged as effective non-traditional machining processes for titanium alloys, enabling the fabrication of complex geometries with high dimensional accuracy and minimal mechanical stresses. This review paper presents a comprehensive analysis of EDM and WEDM processes applied to titanium alloys, focusing on machining mechanisms, influential process parameters, performance characteristics, surface integrity, and metallurgical effects. Various modeling and optimization techniques, including statistical methods, response surface methodology, multi-objective optimization, and artificial intelligence-based approaches, are critically reviewed. The study highlights current advancements, identifies existing research gaps, and outlines future research directions aimed at improving machining efficiency, surface quality, and process sustainability. This review provides a consolidated knowledge base for researchers and industry practitioners involved in advanced machining of titanium alloys.

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1. Introduction

The continuous advancement of modern engineering industries such as aerospace, automotive, biomedical, and defense sectors has led to an increasing demand for materials that combine high strength, low density, excellent corrosion resistance, and superior thermal stability. Titanium and its alloys fulfill these requirements and are therefore extensively used in critical applications including aircraft structural components, turbine parts, medical implants, and high-performance automotive systems. Among various titanium alloys, α–β alloys such as Ti-6Al-4V and near-α alloys like VT-20 (BT-20) are particularly important due to their balanced mechanical properties and thermal stability.

Despite their attractive properties, titanium alloys are classified as difficult-to-machine materials when processed using conventional machining techniques. Poor thermal conductivity, high chemical reactivity at elevated temperatures, strong work hardening tendency, and rapid tool wear significantly limit productivity and dimensional accuracy in traditional machining processes. These challenges necessitate the adoption of non-traditional machining methods capable of machining titanium alloys efficiently without direct mechanical contact.

Electrical Discharge Machining (EDM) and Wire Electrical Discharge Machining (WEDM) have emerged as effective non-traditional machining processes for shaping electrically conductive materials regardless of their hardness. These processes operate on the principle of controlled electrical spark erosion, where material removal occurs through localized melting and vaporization caused by high-temperature plasma generated between the electrode and the workpiece in a dielectric medium. Due to the absence of cutting forces, EDM and WEDM are particularly suitable for machining titanium alloys with complex geometries, thin sections, and tight dimensional tolerances.

Wire Electrical Discharge Machining (WEDM), a specialized variant of EDM, utilizes a continuously moving thin wire electrode to generate controlled spark discharges, enabling precision cutting of intricate two-dimensional and three-dimensional profiles. The process is widely applied in the fabrication of dies, molds, aerospace components, and biomedical devices. However, machining performance in EDM and WEDM is highly sensitive to a large number of process parameters such as pulse on-time, pulse off-time, discharge current, servo voltage, wire tension, wire feed rate, and flushing pressure. Improper selection of these parameters can result in poor surface finish, low material removal rate (MRR), wire breakage, excessive recast layer formation, and reduced dimensional accuracy.

Extensive research has been conducted to investigate the influence of EDM and WEDM parameters on machining performance and surface integrity of titanium alloys. Various statistical, empirical, and optimization techniques—including Taguchi methods, analysis of variance (ANOVA), response surface methodology (RSM), and multi-objective optimization—have been employed to enhance process efficiency. However, the complex thermo-electrical nature of the process, coupled with material-dependent behavior, continues to pose challenges in achieving consistent machining performance.

In this context, the present review paper provides a comprehensive synthesis of published research on EDM and WEDM of titanium alloys. The review focuses on machining mechanisms, parametric effects, surface integrity characteristics, modeling and optimization approaches, and practical industrial applications. Furthermore, critical research gaps and future directions are identified to guide further advancements in EDM and WEDM of titanium alloys

2. **Literature Review**

Titanium alloys such as Ti–6Al–4V and Grade 2 are widely used in aerospace, biomedical, marine and chemical sectors due to their high strength-to-weight ratio, corrosion resistance and temperature capability, but these same properties (low thermal conductivity, high melting point, chemical reactivity) make them difficult-to-cut by conventional machining, leading to high tool wear and low productivity [1] [2] [3] [4] [5] [6]. This has driven extensive adoption of electrical discharge machining (EDM) and wire electrical discharge machining (WEDM) as key non-traditional processes for titanium alloys.

EDM of Titanium Alloys

EDM removes material by repetitive electrical discharges between a tool electrode and a submerged conductive workpiece in a dielectric, generating local temperatures of 6000–20 000 °C that melt and vaporize material without mechanical contact [2] [3] [4]. For titanium alloys, common issues are low material removal rate (MRR), high tool wear rate (TWR), thick recast (white) layer and pronounced heat-affected zone (HAZ), all aggravated by titanium's low thermal conductivity and high melting point [1] [2] [3] [4] [5] [6].

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Fundamental studies have related EDM productivity in Ti–6Al–4V to thermal and electrical properties and to energy partition between tool and workpiece, showing that polarity, pulse duration, and energy density critically control MRR and TWR [3] [5]. Experimental work on Grade 2 titanium and Ti–6Al–4V confirms that pulse current and pulse-on time strongly influence MRR, TWR, and white layer thickness, and that response surface and ANOVA-based models can reliably predict process behavior [3] [7] [5].

To overcome low MRR and high TWR, researchers have explored alternative electrode materials, powder- or surfactant-mixed dielectrics, near-dry EDM, and hybrid processes, often reporting improved flushing, reduced defects and better surface finish [1] [4] [8] [6]. Near-dry EDM milling of Ti–6Al–4V, for example, uses a gas–liquid mixture and multi-objective optimization (GRA) to identify parameter sets that jointly enhance MRR, surface quality and dimensional accuracy [9].

Wire EDM (WEDM) of Titanium Alloys

WEDM uses a moving thin wire (typically 0.05–0.3 mm) as the electrode to produce intricate 2D/3D contours with high dimensional accuracy [10] [11] [12] [13]. For titanium alloys, WEDM is increasingly used in aerospace components and biomedical implants, where tight tolerances and surface integrity are critical [10] [14] [11] [15] [8].

A central limitation in WEDM of titanium is wire rupture and instability under severe conditions, linked to high cutting temperature, poor heat dissipation and debris accumulation in the narrow spark gap [1] [11] [15] [13]. Studies on Ti–6Al–4V show that pulse-on time, wire tension and flushing pressure significantly affect kerf width, discharge gap, MRR, surface roughness and wire degradation, with multi-layered recast layers containing cracks, pores and wire-material transfer commonly observed [1] [3] [11] [13]. Recast layers often include TiC and TiO_2 phases, depending on the dielectric and discharge conditions, which can alter hardness and corrosion behavior [1] [3] [4].

Recent WEDM-specific reviews and experiments emphasize the importance of process optimization and modelling. Taguchi, Grey Relational Analysis, hybrid Taguchi–Grey and response surface methods have been widely used to optimize MRR, surface roughness, kerf width, dimensional errors, microhardness, recast-layer thickness and wire wear ratio [16] [14] [12] [15] [17] [8]. Multi-objective frameworks and, more recently, machine-learning models (e.g., random forests, LightGBM) have been applied to predict surface roughness, crack density, kerf width and corner accuracy for Ti-6Al-4V WEDM, enabling parameter selection with high predictive accuracy [18] [19] [17] [20].

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Table 1: Representative EDM/WEDM Performance Factors in Titanium Alloys

Aspect / Challenge	Typical Findings in Ti Alloys	Citations
MRR & TWR in die-sinking EDM	Generally low MRR and high TWR; highly sensitive to pulse current, Ton, polarity and dielectric composition	[1] [2] [3] [4] [7] [5] [6]
Surface integrity (EDM & WEDM)	Thick recast/white layer, HAZ, microcracks, gas pores; TiC/TiO_2 formation; altered microhardness	[1] [2] [3] [4] [11] [5] [8]
Wire rupture (WEDM)	Prominent at high current and long Ton; governed by wire material, tension, flushing and thermal load	[1] [11] [15] [13]
Optimization tools	Taguchi, RSM, ANN, Grey, TGRA, hybrid Taguchi–Grey, and ML models applied to Ti EDM/WEDM	[16] [3] [4] [18] [19] [12] [7] [17] [8] [20]
Bio-/dental applications	WEDM used to machine and texture bio-titanium; surface quality and recast control are critical	[10] [14] [8]

Research Trends and Gaps

Recent reviews on EDM and WEDM of titanium identify broad progress in process understanding, parameter optimization, dielectric and electrode design, and hybrid/near-dry variants, yet also highlight persistent gaps: limited integration of surface integrity with fatigue/corrosion behavior, incomplete models of wire failure in WEDM, and a need for greener dielectrics and in-process monitoring for titanium-specific conditions

Overall, EDM and WEDM have become indispensable for machining titanium alloys, but achieving high productivity with controlled surface integrity remains an active research frontier.

3. Titanium Alloys: Characteristics and Machinability

Titanium alloys occupy a critical position in advanced engineering applications owing to their exceptional combination of mechanical, chemical, and thermal properties. These alloys exhibit a high strength-to-weight ratio, excellent corrosion resistance, superior fatigue behavior, and good biocompatibility, making them indispensable in aerospace, automotive, marine, chemical processing, and biomedical industries. Despite these advantages, titanium alloys present significant challenges during machining, particularly when conventional material removal processes are employed.

3.1 Metallurgical Characteristics of Titanium Alloys

Pure titanium undergoes an allotropic phase transformation at approximately 882 °C, changing from a hexagonal close-packed (α) structure at lower temperatures to a body-centered cubic (β) structure at higher temperatures. Alloying elements are added to stabilize either the α or β phase or to create a mixed α–β microstructure, thereby tailoring mechanical properties and thermal stability.

Based on their microstructural constitution, titanium alloys are generally classified into four categories:

(i) **α alloys**,

(ii) **near-α alloys**,

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(iii) **α–β alloys**, and
(iv) **β alloys**.
Among these, **α–β alloys such as Ti-6Al-4V** and near-α alloys like **VT-20 (BT-20)** are the most widely used in structural and high-temperature applications due to their balanced strength, toughness, and creep resistance. The presence of aluminum acts as an α-stabilizer, while elements such as vanadium and molybdenum stabilize the β phase, resulting in improved mechanical performance.

3.2 Mechanical and Thermal Properties

Titanium alloys demonstrate high tensile strength comparable to alloy steels while being approximately 40–45 % lighter in weight. However, their low thermal conductivity (nearly one-sixth that of steel) leads to rapid heat accumulation in the cutting zone during machining. Additionally, titanium alloys retain high strength at elevated temperatures, which further increases resistance to material removal.

Another important characteristic is their high chemical reactivity at elevated temperatures, causing strong adhesion between the tool and workpiece during conventional machining. This leads to accelerated tool wear, built-up edge formation, and degradation of surface integrity.

3.3 Machinability Challenges of Titanium Alloys

Titanium alloys are widely regarded as difficult-to-machine materials due to the combined effects of thermal, mechanical, and chemical factors. Major machinability challenges include:

- Rapid tool wear caused by diffusion and adhesion at high temperatures
- Poor heat dissipation, leading to localized thermal damage
- Work hardening and high cutting forces
- Low modulus of elasticity, resulting in vibration and dimensional inaccuracies
- Formation of built-up edges and surface damage

These issues significantly reduce tool life, machining accuracy, and productivity in conventional machining processes such as turning, milling, and drilling.

3.4 Suitability of EDM and WEDM for Titanium Alloys

Electrical Discharge Machining (EDM) and Wire Electrical Discharge Machining (WEDM) overcome many of the limitations associated with conventional machining of titanium alloys. Since these processes rely on **electro-thermal erosion rather than mechanical cutting**, material hardness and strength have minimal influence on machinability.

Key advantages of EDM and WEDM in titanium alloy machining include:

- Absence of mechanical cutting forces
- Capability to machine hardened and heat-treated alloys
- High dimensional accuracy and repeatability
- Ability to produce complex and intricate geometries
- Reduced residual stresses compared to conventional machining

However, the high thermal energy involved in EDM and WEDM can result in recast layers, heat-affected zones (HAZ), micro-cracks, and surface oxidation, particularly in titanium alloys. Therefore, careful selection and optimization of process parameters are essential to balance material removal rate, surface integrity, and dimensional accuracy.

3.5 Summary

In summary, titanium alloys possess outstanding mechanical and chemical properties that make them ideal for critical engineering applications, but these same properties severely limit their machinability using traditional methods. EDM and WEDM provide effective alternatives for machining titanium alloys, particularly for complex and precision components. A thorough

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understanding of the material characteristics and their interaction with EDM/WEDM process parameters is essential for improving machining performance and ensuring surface integrity.

4. Modeling and Optimization Techniques

The complex thermo-electrical nature of Electrical Discharge Machining (EDM) and Wire Electrical Discharge Machining (WEDM) necessitates the use of robust modeling and optimization techniques to predict machining performance and determine optimal process conditions. Machining responses such as material removal rate (MRR), surface roughness (Ra), kerf width, dimensional accuracy, and wire breakage are strongly influenced by multiple interacting parameters. Consequently, researchers have employed a wide range of statistical, empirical, semi-empirical, and computational approaches to model and optimize EDM and WEDM processes for titanium alloys.

4.1 Statistical and Empirical Modeling Approaches

Statistical modeling techniques are widely used due to their simplicity, efficiency, and ability to capture parametric effects with a limited number of experiments. Among these, the Taguchi method has been extensively applied to identify significant machining parameters and their optimal levels. By employing orthogonal arrays and signal-to-noise (S/N) ratios, Taguchi-based studies have effectively analyzed the influence of pulse on-time, pulse off-time, peak current, and servo voltage on MRR and surface roughness in titanium alloy machining.

Analysis of Variance (ANOVA) is commonly used in conjunction with Taguchi designs to quantify the percentage contribution of each parameter and to determine statistically significant factors. Numerous studies report that pulse on-time and discharge current are the most dominant parameters affecting MRR and surface integrity, while pulse off-time and servo voltage significantly influence machining stability and surface finish.

Regression-based empirical models have also been developed to establish mathematical relationships between input parameters and output responses. These models provide useful predictive capability within the experimental domain but are generally limited to specific materials, machines, and parameter ranges.

4.2 Response Surface Methodology (RSM)

Response Surface Methodology (RSM) is one of the most widely adopted optimization techniques in EDM and WEDM research. RSM employs second-order polynomial equations to model the relationship between multiple process variables and performance responses. Central Composite Design (CCD) and Box–Behnken Design (BBD) are commonly used experimental designs in RSM studies.

For machining titanium alloys, RSM has been successfully applied to:

- Predict material removal rate and surface roughness
- Analyze interaction effects among process parameters
- Perform multi-objective optimization

RSM offers higher prediction accuracy compared to single-factor or Taguchi methods and enables visualization of response surfaces and contour plots. However, the accuracy of RSM models decreases outside the experimental domain, highlighting the need for validation experiments.

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4.3 Multi-Response Optimization Techniques

Since EDM and WEDM processes often involve conflicting objectives (e.g., high MRR and low surface roughness), multi-response optimization techniques have gained significant attention. **Grey Relational Analysis (GRA)** is widely used to convert multiple responses into a single grey relational grade, enabling simultaneous optimization of performance characteristics.

Other multi-objective optimization methods include:

- Pareto optimization
- Desirability function approach
- Weighted sum methods

These techniques provide a set of optimal solutions rather than a single optimal point, allowing flexibility in selecting machining conditions based on application-specific requirements.

4.4 Semi-Empirical and Physics-Based Models

Semi-empirical models attempt to integrate machining parameters with thermo-physical properties of the workpiece material, such as melting point, thermal conductivity, and coefficient of thermal expansion. Dimensional analysis combined with non-linear regression techniques has been used to develop such models for predicting MRR in WEDM of titanium alloys and metal matrix composites.

Although physics-based models offer deeper insight into the material removal mechanism, they often involve simplifying assumptions related to plasma channel behavior, spark energy distribution, and debris flushing, leading to discrepancies between predicted and experimental results.

4.5 Artificial Intelligence and Machine Learning Approaches

Recent advancements in computational intelligence have introduced artificial neural networks (ANNs), fuzzy logic, genetic algorithms (GA), and hybrid AI models for EDM and WEDM optimization. These techniques are capable of handling non-linear relationships and complex parameter interactions more effectively than traditional statistical models.

Machine learning-based models have demonstrated high prediction accuracy for machining responses such as MRR, surface roughness, and kerf width. Genetic algorithms and other evolutionary techniques have been used to optimize process parameters by coupling them with ANN or RSM models.

Despite their superior performance, AI-based approaches require large experimental datasets and computational resources, which may limit their industrial applicability.

4.6 Summary

Modeling and optimization techniques play a crucial role in enhancing the performance and reliability of EDM and WEDM processes for titanium alloy machining. While traditional statistical methods such as Taguchi and RSM remain popular due to their simplicity and effectiveness, advanced approaches including multi-objective optimization and machine learning offer promising avenues for improved prediction accuracy and adaptive control. Future research should focus on integrating physics-based models with data-driven techniques to develop robust, scalable, and industry-ready optimization frameworks.

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5. Conclusion

This review paper has presented a comprehensive and critical overview of Electrical Discharge Machining (EDM) and Wire Electrical Discharge Machining (WEDM) with specific emphasis on the machining of titanium alloys. Owing to their superior mechanical properties, corrosion resistance, and high temperature stability, titanium alloys are widely used in advanced engineering applications; however, their poor machinability using conventional methods necessitates the adoption of non-traditional machining techniques.

The fundamental principles of EDM and WEDM, material removal mechanisms, and the role of thermo-electrical interactions were discussed to provide a clear understanding of process behavior. A detailed analysis of machining parameters revealed that pulse on-time, discharge current, pulse off-time, and servo voltage are the most influential factors governing material removal rate, surface roughness, dimensional accuracy, and machining stability. The review further highlighted the significant impact of wire material, flushing conditions, and electrical energy input on surface integrity, including recast layer formation, heat-affected zone development, and microstructural alterations in titanium alloys.

Modeling and optimization techniques reviewed in this paper demonstrate a progressive evolution from traditional statistical approaches such as Taguchi methods and ANOVA to advanced techniques including response surface methodology, multi-objective optimization, and artificial intelligence-based models. While statistical and empirical models offer simplicity and cost-effectiveness, advanced machine learning and hybrid optimization techniques provide improved prediction accuracy and adaptability for complex, non-linear machining behavior. Nevertheless, their industrial implementation remains limited due to data dependency and computational requirements.

Despite extensive research efforts, several challenges remain unresolved, particularly in achieving an optimal balance between high productivity and superior surface integrity. Future research should focus on the integration of physics-based modeling with data-driven approaches, real-time adaptive control systems, sustainable machining strategies, and deeper investigation into metallurgical transformations during EDM and WEDM of titanium alloys.

Overall, this review provides a consolidated knowledge base for researchers and practitioners, offering valuable insights into current developments, limitations, and future prospects of EDM and WEDM processes for titanium alloy machining.

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